

KAVALI SOFIA SAGAR

PhD Student at Indian Institute of Technology Hyderabad

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EDUCATION

B.Tech Computer Science and Engineering

Indian Institute of Information Technology Kottayam

August 2020 – June 2024 Kottayam, Kerala, India

- Relevant Courses: Artificial Intelligence, Operating Systems, Theory of Computation, Object-Oriented Analysis and Design, Compiler Design, Soft Computing, Cloud Computing, Blockchain Technology
- Graduated with Honors, **95.8%** cumulative average

B.S. in Data Science and Applications (Online)

Indian Institute of Technology Madras

Jan 2021 – Mar 2025

- Relevant Courses: Privacy and security in online social media, Modern Application Development, AI: search methods for problem solving, Database Management Systems, Data Structures and Algorithms
- Graduated with **8.04** cumulative average

PhD Artificial Intelligence

Indian Institute of Technology Hyderabad

Aug 2024 - Hyderabad, Telangana, India

- PhD Advisor: **Dr. P. N. Karthik**
- Relevant Courses: Probability and Stochastic Processes, Advanced Data Structures and Algorithms, Foundations of Machine Learning, Stochastic Processes: Theory and Applications, Deep Learning
- Cumulative Grade Point Average: **8.75**

WORK EXPERIENCE

Centre for Development of Advanced Computing, India

June 2024 - July 2024 Bengaluru, Karnataka, India

- Gained initial exposure to applied Research and Development workflows and disciplined software practices.

SKILLS

PythonPytorchOpenCVC++CCJavaHTMLCSS

JavaScriptVue JSDBMSPostgre SQLRPHPLatex

ACHIEVEMENTS

- GATE (Data Science & Artificial Intelligence) – All India Rank **336**

RESPONSIBILITY

Indian Institute of Technology Hyderabad

Aug 2025 - Present

- Teaching Assistant: Probability And Stochastic Processes (Aug-2025)
- Teaching Assistant: Statistical Learning Theory (Aug-2025)

PROJECTS

Human-Centric Vision & Interaction

Real-Time Indian Sign Language (ISL) Detection and Touch-less Keyboard

- Built a real-time web interface for hand-motion recognition using MediaPipe, classifying letters, digits, words, and dynamic actions.
- Lightweight descriptors for low-compute environments and implemented a text-to-sign module with touch-less keyboard to enhance accessibility.

Systems & Cryptography

Secure Touch-less Voting with Blockchain

- Prototyped a camera-based, contactless voting interface using MediaPipe, integrated with a blockchain backend for auditability and tamper-evidence.
- Implemented multiple tallying methods (e.g., Borda count, single transferable vote) and automated statistical graphs.

Web Systems & Machine Learning

Movie Ticketing & Review Classification

- Developed a movie ticketing system with admin/user workflows using Flask, SQLite, Celery (backend) and Vue.js (frontend), ensuring reliable transactions and responsive UI.
- Built a movie review classifier by implementing and comparing classical ML models (SVM, Decision Trees, Perceptron), analyzing trade-offs between accuracy, model complexity, and feature choices.

Data Science & Analytics

Data Visualization & Analysis Projects

- Applied a full toolkit for data extraction, cleaning, visualization, and dash-boarding (Power BI, Figma, and related tools)
- Analyzed NSE stock data to identify historical trends, built explanatory visualizations, and tested forecasting baselines with error evaluation.

LANGUAGES

English

Telugu

Hindi

French